

Definition of Product Design

- Product design is the process of imagining, creating, and iterating products that solve users' problems or address specific needs in a given market.
- The key to successful product design is understanding the end-user customer, the person for whom the product is being created.
- Product design is one of the essential steps in the Operations Management structure. It is one of the necessary formulas to achieve the ultimate goal of consumer satisfaction.

Process of Product Design

In general, there are 5 main product design phases:

1. Discussing plans for launching a new product within the team, brainstorming;
2. Defining pain points (desires) of the consumer and solutions for their elimination (achievement);
3. Developing strict product requirements (documenting technical specifications);
4. Splitting the product implementation process into iterations;
5. Testing and modifying the created solution on the basis of real usage and target user experience.

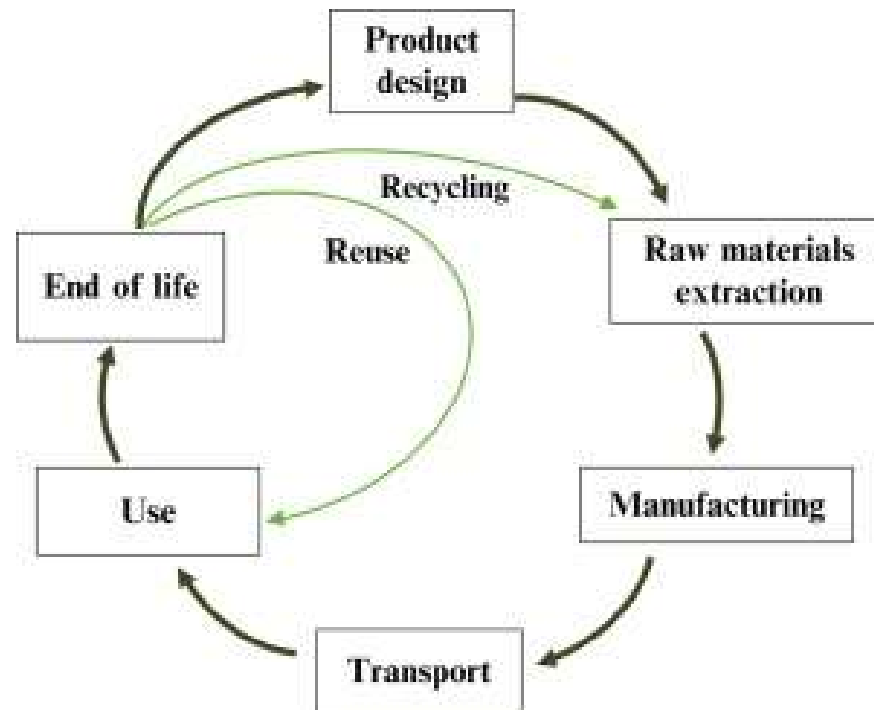
Ten Steps of the Product Design Process

- ❖ Brainstorming
- ❖ Defining the Product
- ❖ Conducting the User Research
- ❖ Sketching
- ❖ Prototyping
- ❖ Compiling Specifications
- ❖ Producing the Factory Samples
- ❖ Sample Testing
- ❖ Starting the Production/Development
- ❖ Providing Quality Assurance

Guiding Principles of Product Design

- I. Utility
- II. Aesthetic Appeal
- III. Simplicity
- IV. Innovation
- V. Longevity

Consumption Cycle and Product Life Cycle



Product Life Cycle

- A product life cycle is the length of time from a product first being introduced to consumers until it is removed from the market.
- A product's life cycle is usually broken down into four stages; introduction, growth, maturity, and decline.
- Product life cycles are used by management and marketing professionals to help determine advertising schedules, price points, expansion to new product markets, packaging redesigns, and more.
- These strategic methods of supporting a product are known as product life cycle management.

Product Life Cycle

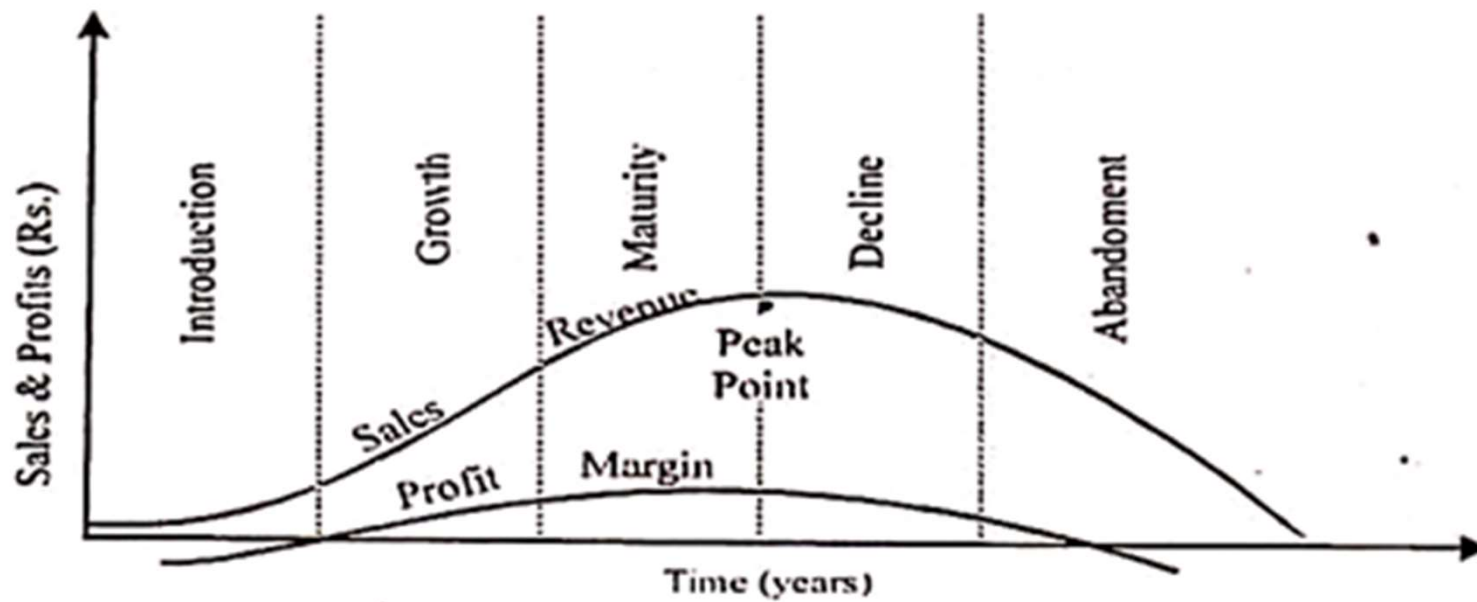
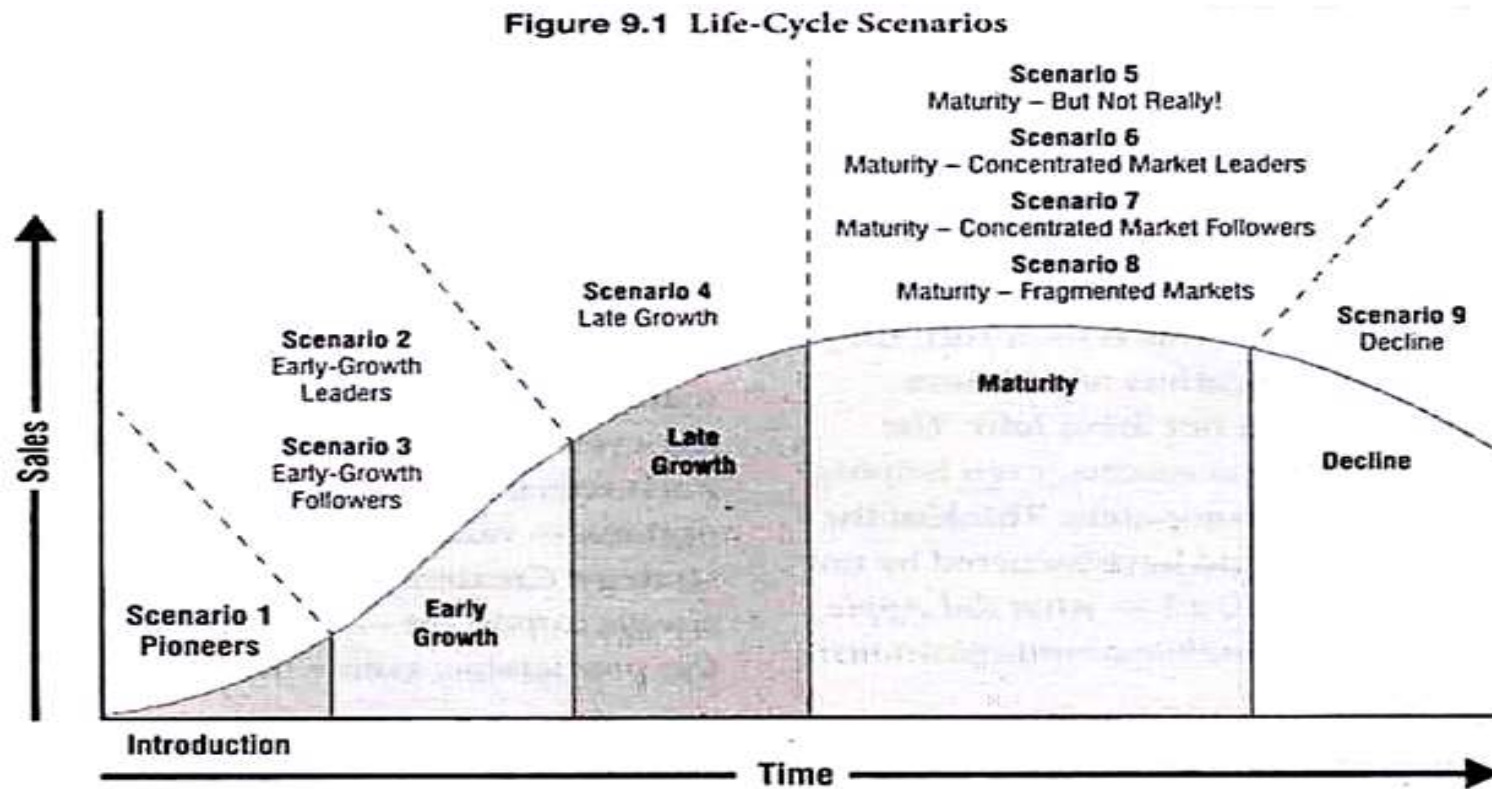
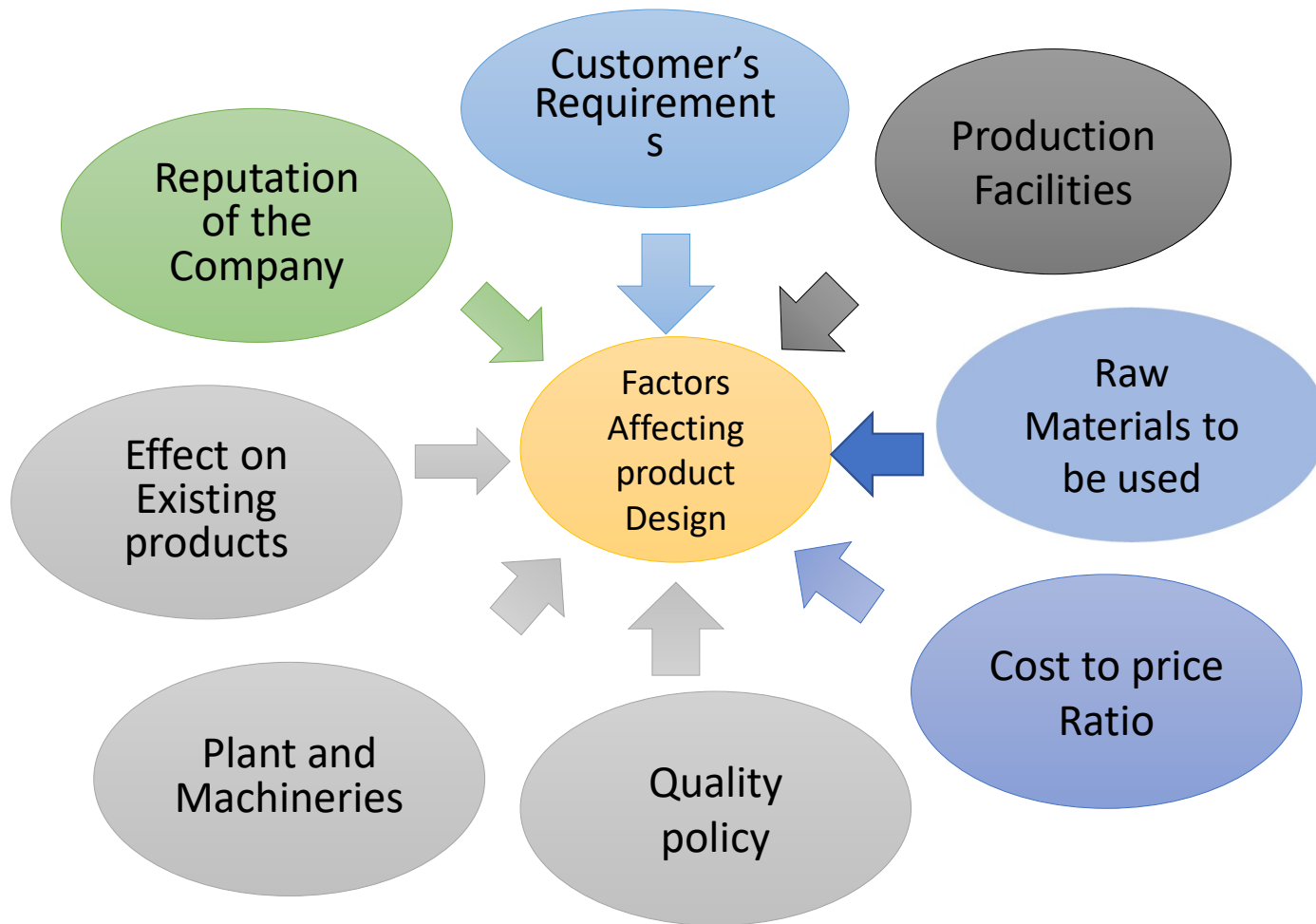


Fig. 15.2 : Product Life-Cycle

Product Life Cycle Scenarios



Factors Affecting Product Design



Simplification in material Management

- The concept of simplification is closely related to standardization.
- Simplification is the process of reducing the variety of products manufactured
- Simplification is concerned with the reduction of products range, assemblies, parts, materials and design.
- Advantages of simplifications
 - Involves fewer, parts, varieties and changes in products, this reduces manufacturing operations and risk of obsolescence.
 - Reduces variety; volume of remaining products may be increased.
 - Provides quick delivery and better after sales services.
 - Reduces inventory
 - Lowers the product cost
 - Reduces price of a product.
 - Improve product quality.

Standardization

- The standardization process is where all the parties belonging to a particular industry or sharing common interests come together to form an agreement defining standards and guidelines to be followed.
- Different stages of the process include selection, codification, elaboration, and acceptance.
- It creates uniformity or consistency for specific actions or procedures inside the selected setting.
- It has wide applications in trading stocks and commodities, accounting, manufacturing, management, judicial system, etc.

The standardization process involves the following stages:

Selection: This stage involves choosing or defining the standards.

Codification: This stage involves codifying the standards selected for the people or stakeholders involved to follow

Elaboration: It is the extension or improvisation of standards for implementation in new domains.

Acceptance: It involves the acceptance of selected standards by strictly following, promoting, and establishing them.

Advantages of Standardization

- **Fewer parts** to deal with in inventory & manufacturing
 - Less costly to fill orders from inventory
- **Reduced training** costs and time
- More **routine** purchasing, handling, and inspection procedures
- Opportunities for long production runs, **automation**
- Need for fewer parts justifies increased expenditures on perfecting designs and improving quality control procedures.
- Decreased variety results in **less consumer appeal**.
- Designs may be frozen with **too many imperfections** remaining.
- High **cost of design changes** increases resistance to improvements
- Standard systems are **more vulnerable** to failure

Effect of Standardization

- It helps bring a smooth workflow, improves end product quality, and focuses on consistency.
- Business standards can help innovation and act quickly in changing environment.
- Companies apply it to study their competitors and develop continuous improvement in their final deliverance.
- One of the major effects that customers experience is consistency in products, interoperability, and compatibility.
- It brings technological advancement whenever necessary to procure better services and gain a new competitive advantage in the market.
- One of the key effects of it associated with mass production is the strict adherence to protocols and defined systems.
- If not given attention, it can affect the quality and reduces healthy competition.

SPECILISATION

- Product specialization is a marketing technique where businesses focus their marketing or branding efforts on a specific product or product line.
- These marketing efforts often focus on the benefits and quality of the product that may attract potential customers.
- Companies use this technique to target a specific market instead of the general public.
- Some companies that use product specialization build their reputation as experts in the specific market which they're targeting, meaning their products become the standard for that specific market

SPECILISATION

Benefits of product specialization

- Market concentration

Product specialization allows companies to focus on one or two markets for their products, reducing marketing efforts to target specific customers instead of the mass public. This may reduce advertising costs.

- Competition reduction

Specific markets usually have a set amount of companies that operate within them, which often means fewer competitors

- After-sale product sales

Consumers who buy these specialized products may be more likely to buy more products that enhance their purchases from the company

- Increased focus on product-specific ads:

A combination of market concentration and reduced competition means some companies can dedicate marketing efforts to developing specific ad campaigns

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Inter-changeability

- Interchangeability means a product is interchangeable with another if both products are approved for the same indication, and can be used for the said indication.
- Product interchangeability in supply network planning is used to transfer demand of a product to be discontinued on to a successor product or to use existing stock of a current product to meet demand for the successor product.
- **Possible business scenarios**
- A product is planned to be replaced with a technically improved product.
- A defective product needs to be discontinued and switched to a non-defective, good quality product
- There are multiple products which are similar in their technical properties and can be substituted with one another
- Temporary replacement of a product to promote another product
- **Product interchangeability function helps the organizations to achieve the following:**
- Better discontinuation planning and execution
- Optimizing material inventory and availability
- Minimizing scrap